

CT details.

1. Compressor current (measured): 500mA (rms)
2. Number of primary turns: 2
3. Current in the secondary: 1mA
4. Voltage across burden R1: 4.7V (rms)
5. Peak to peak voltage across R1: $2\sqrt{2} \times 4.7 = 13.3\text{Vpp}$

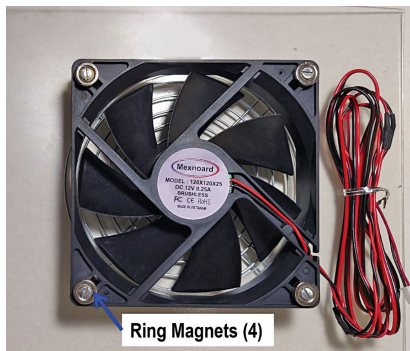


Fig. 5: Front view of fan assembly with four magnets

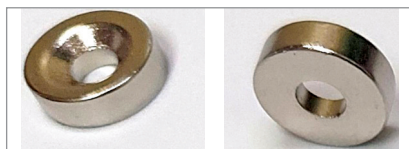


Fig. 6: Close-up view of ring magnets

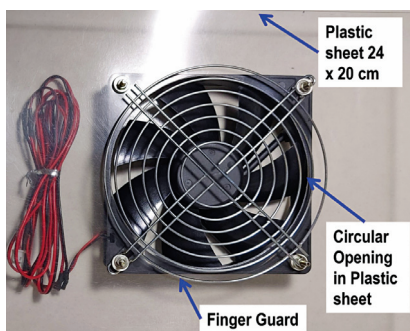


Fig. 7: Back-side view of the fan assembly

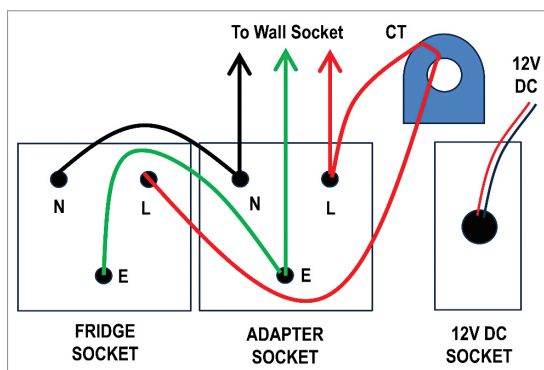


Fig. 8: Wiring diagram of switch box

Fig. 2 shows the starting current waveform of the compressor (CT burden $4.7\text{k}\Omega$, primary 2 turns), while Fig. 3 shows the running current waveform. The CT output is rectified using a full-wave rectifier consisting of diodes D1 through D4 (1N4007). The rectified voltage is filtered using C1 ($0.47\mu\text{F}$) box capacitor. This rectified current signal drives transistor T1 (BC546) through resistor R2 ($4.7\text{k}\Omega$). T1 drives the power transistor T2 (TIP31C) using current-limiting resistor R3 (330Ω , 1W). T2 is mounted on a small heat-sink. In the collector circuit of T2, two 12V instrument cooling fans are connected. D5 (1N4007) acts as a free-wheeling diode and eliminates voltage overshoot whenever the fans turn off.

Construction

The assembled prototype of the system is shown in Fig. 4. Polarised 2-pin connectors CON1 and CON2 are used to connect the fans. CON3 (terminal block) is used to connect the +12V and RET wires coming from the DC jack. An adaptor (12V, 1A) is used to power the circuit.

The PCB layout is shown in Fig. 4. This PCB is mounted inside a switch box using the two mounting holes.

Fan assembly

Low-noise cooling fans are selected. A larger $120\text{mm} \times 120\text{mm}$ fan helps reduce noise while providing a bigger cooling area.

Fig. 5 shows the front view of the fan assembly. Four ring magnets (12mm dia $\times$ 4mm thick) are fixed at the four mounting holes of the fan using $\text{M3} \times 40\text{mm}$ screws. A close-up view of the magnets is shown in Fig. 6.

The screws are passed through the holes of the magnet and fixed as shown. The magnets create a 4mm gap between the the fridge surface and the

fan blades. If more gap is required for smoother airflow, extra nylon washers can be added between the magnet and the fan housing.

This assembly can be magnetically attached at any desired location on the fridge side walls. Do not drag the fan on the fridge surface, as it may scratch the paint.

Fig. 7 shows the back-side view. A stiff plastic sheet of about $240\text{mm} \times 200\text{mm}$ is fixed as shown. A circular opening of the same size as the fan blade diameter is cut in the plastic sheet to allow airflow.

A finger guard is then placed, and the nuts are tightened. The plastic sheet redirects airflow from the fan onto the fridge surface, thus increasing the effective cooling area.

Fan power consumption.

1. Fan operating voltage: 12V DC
2. Fan operating current: 125mA (measured)
3. Power consumed: $12 \times 0.125 = 1.5\text{W}$
4. No of fans: 2
5. Total power consumed: $2 \times 1.5 = 3\text{W}$
6. Compressor duty cycle: 50%
7. Energy consumed in 24 hrs: $24 \times 0.5 \times 3 = 36\text{Wh}$
8. Energy consumed/month: $36 \times 30 = 1080\text{Wh}$ (about one unit)
9. Running cost/month: ₹5

Wiring and assembly. Fig. 8 shows the wiring diagram. Two 5A sockets are used here, one to power the adaptor and the other to power the fridge.

A short cable with a 3-pin plug is used to get power from the wall outlet where the refrigerator is currently plugged in. The phase wire is first passed through the CT and then connected to the socket meant for the fridge. The actual wiring is shown in Fig. 9.

Caution. Both 230V and 12V supplies are present in the same enclosure. Therefore, extra care must be taken while wiring.

Use good-quality sockets in which the terminals are not exposed. It is recommended to cover the sockets